Assignment - 9

1) Hypothesis

$H_0 : \mu_A = \mu_B = \mu_C = \mu_D$

H_1 : At least one mean is different.

Given data

Brand A	1610	1610	1650	1680	1700	1720	1800	
Brand B	1580	1640	1640	1700	1750			
Brand C	1460	1550	1600	1620	1640	1660	1740	1820
Brand D	1510	1520	1530	1570	1600	1680		

Total :-

$T_A = 11770, \; n_A = 7$

$T_B = 8310, \; n_B = 5$

$T_C = 13090, \; n_C = 8$

$T_D = 9410, \; n_D = 6$

$T = 42580, \; N = 26.$

Means.

$\bar{X}_A = 1681.43, \; \bar{X}_B = 1662, \; \bar{X}_C = 1636.25, \; \bar{X}_D = 1568.33$

Correction Factor :-

$$CF = \frac{T^2}{N} = \frac{(42580)^2}{26} = 69740307.69.$$

Sum of Squares:

$$SST = \sum x^2 - CF = 70092200 - 69740307.69$$
$$= 351892.31$$

Between groups (SSB)

$$SSB = \frac{11770^2}{7} + \frac{8310^2}{5} + \frac{13090^2}{8} + \frac{9410^2}{6} - CF$$
$$= 69903023.57 - 69740307.69$$
$$= 162715.88.$$

Error (SSE)

$$SSE = 351892.31 - 162715.88 = 189176.43$$

ANOVA Table

Source	SS	df	MS	F
Between (Brands)	162715.88	3	54238.63	6.31
Within (Error)	189176.43	22	8598.02	
Total	351892.31	25		

F-Test

$$F = \frac{54238.63}{8598.02} = 6.31$$

Table Value :-

$$F_{(3,22)} \approx 3.05.$$

$$6.31 > 3.05$$

Conclusion :-

Ho is rejected

The mean lives of the four brands differ Significantly.

2) Hypothesis

$H_0: \mu_A = \mu_B = \mu_C$

H_1: At least one mean differs.

Given data :

A	B	C
20	18	25
21	20	28
23	17	22
16	25	28
20	15	32.

Total

$T_A = 100$, $T_B = 95$, $T_C = 135$

$n = 5$ (each) , $N = 15$.

$T = 330$

Correction Factor

$$CF = \frac{T^2}{N} = \frac{330^2}{15} = 7260$$

Sum of Squares

$$SST = \sum x^2 - CF$$

$$= (400+441+529+256+400)+(324+400+289+625+225)$$
$$+(625+784+484+784+1024)$$

$$= 2026 + 1863 + 3701$$

$$= 7590$$

$$SST = 7590 - 7260 = 330$$

Treatment SS

$$SSTR = \frac{100^2}{5} + \frac{95^2}{5} + \frac{135^2}{5} - 7260$$

$$= 2000 + 1805 + 3645 - 7260$$

$$= 190.$$

Error SS

$$SSE = 330 - 190 = 140$$

ANOVA Table.

Source	SS	df	MS	F
Between (Varieties)	190	2	95	8.14.
Within (Error)	140	12	11.67	
Total	330	14.		

F-Test

$$F = \frac{95}{11.67} = 8.14.$$

Table Value

$$F_{(2,12)} \approx 3.89.$$

Conclusion:-

$$8.14 > 3.89$$

Reject H_0.

The yields of the three seed varieties differ significantly

3) Hypothesis

H_0: All technicians are equally efficient.

H_1: At least one technician differs

For days

H_0: No difference between days

H_1: Difference exists between days.

Given data

Technician I (x_1)	Technician II (x_2)	Technician III (x_3)	Technician IV (x_4)
6	14	10	9
14	9	12	12
10	12	7	8
8	16	15	10
11	14	11	11

Column Totals

$C_1 = 49$, $C_2 = 59$, $C_3 = 55$, $C_4 = 50$

Grand Total

$T = 213$, $N = 20$

Correction Factor

$$CF = \frac{T^2}{N} = \frac{213^2}{20} = 2268.45$$

Total Sum of Squares

$$SST = \sum x^2 - CF$$

$$\sum x^2 = 2457$$

$$SST = 2457 - 2268.45 = 188.55$$

Row Sum of Squares (Technicians)

$$SSR = \frac{39^2+47^2+37^2+43^2+47^2}{4} - CF$$

$$= \frac{1521+2209+1369+1849+2209}{4} - 2268.45$$

$$= 2281.40 - 2268.45 = 20.80$$

Column Sum of Squares (Days)

$$SSC = \frac{49^2+59^2+55^2+50^2}{5} - CF$$

$$= \frac{2401+3481+3025+2500}{5} - 2268.45$$

$$= 2281.40 - 2268.45 = 12.95$$

Error

$$SSE = 188.55 - 20.80 - 12.95 = 154.80.$$

ANOVA Table.

Source	SS	df	MS	F
Technicians	20.80	4	5.20	0.40
Days	12.95	3	4.32	0.33
Error	154.80	12	12.90	
Total	188.55	19		

F-Test

$$F_{Tech} = \frac{5.20}{12.90} = 0.40$$

$$F_{Days} = \frac{4.32}{12.90} = 0.33.$$

Conclusion.

CCV)
$0.40 < 3.26$ (TV)

accept Ho

There is no significant difference among technicians

There is no significant difference among days.

4) Hypothesis

for Treatment

H_0: All treatments have equal effect

H_1: At least one treatment differs.

for blocks.

H_0: No block effect

H_1: Block effect exists.

Given data:

		Treatments.			
		T_1	T_2	T_3	T_4.
Blocks.	B_1	12	14	20	22
	B_2	17	27	19	15
	B_3	15	14	17	12
	B_4	18	16	22	12
	B_5	19	15	20	14

Column Total.

$T_1 = 81$, $T_2 = 86$, $T_3 = 98$, $T_4 = 75$

Grand Total

$T = 340$, $N = 20$

Correction Factor

$$CF = \frac{T^2}{N} = \frac{340^2}{20} = 5780$$

Total Sum of Squares

$$\Sigma x^2 = 6064.$$

$$SST = 6064 - 5780 = 284.$$

Treatment Sum of Squares

$$SSTr = \frac{81^2 + 86^2 + 98^2 + 75^2}{5} - 5780.$$

$$= 5832.4 - 5780 = 52.4.$$

Block Sum of Squares.

$$SSB = \frac{68^2 + 78^2 + 58^2 + 68^2 + 68^2}{4} - 5780$$

$$= 5830 - 5780 = 50$$

Error

$$SSE = 284 - 52.4 - 50 = 181.6$$

ANOVA Table.

Source	SS	df	MS	F
Treatments	52.4	3	17.47	1.44
Blocks	50	4	12.5	1.03
Error	181.6	12	15.13	
Total	284	19		

F-Test

$$F_{Treatment} = 1.44.$$

$$F_{Block} = 1.03$$

Table Values:

$$F(3,12) \approx 3.49, \quad F(4,12) \approx 3.26.$$

Conclusion:

Since Calculated F-Values are less than table Values, we accept H_0

There is no significant difference among treatments.

There is no significant difference among blocks.

5) Given:-

The regression equation between X & Y & estimate required value

Data

	1	2	3	4
1	A(12)	B(20)	C(16)	B(10)
2	D(18)	A(14)	B(11)	C(14)
3	B(12)	C(15)	D(19)	A(13)
4	C(16)	B(11)	A(15)	D(20)

X	Y
10	12
12	15
13	17
16	20
18	22

Calculate mean

$$\bar{X} = \frac{10+12+13+16+18}{5} = 13.8$$

$$\bar{Y} = \frac{12+15+17+20+22}{5} = 17.2$$

Table Calculation

X	Y	$X-\bar{X}$	$Y-\bar{Y}$	$(X-\bar{X})^2$	$(X-\bar{X})(Y-\bar{Y})$
10	12	-3.8	-5.2	14.44	19.76
12	15	-1.8	-2.2	3.24	3.96
13	17	-0.8	-0.2	0.64	0.16
16	20	2.2	2.8	4.84	6.16
18	22	4.2	4.8	17.64	20.16

$$\Sigma(X-\bar{Y})^2 = 40.8$$

$$\Sigma(X-\bar{X})(Y-\bar{Y}) = 50.2.$$

Regression Coefficient

$$b_{yx} = \frac{50.2}{40.8} = 1.23$$

Regression Equation

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$y - 17.2 = 1.23(x - 13.8)$$

$$y = 1.23x + 0.23$$

Final

Regression equation of Y on X:

$$y = 1.23x + 0.23$$

Conclusion

There is a positive relationship between X & Y.